

Solving by completing the square

HANDOUT



Interactive version

GOAL

In this lesson

Rewrite any quadratic equation as $(x + p)^2 = d$ and solve it.

BY THE END OF THE LESSON

- ☐ I can rewrite $x^2 + bx + c$ as $(x + b/2)^2 + q$, also when $b/2$ is a fraction.
- ☐ I can solve a quadratic equation by completing the square, and tell from $(x + p)^2 = d$ whether it has two roots, one root or none.

WHAT YOU NEED FROM BEFORE

- ↔ Lesson 1.1 · Quadratic equations: what we can already solve
- ↔ Operations with fractions
- ↔ Completing the square in simple cases

1 class period • about 45 min

HOOK

Backwards from (E)

Lesson 1.1 ended with two faces of one equation:

- (E) $x^2 - 4x - 5 = 0$
- (D) $(x - 2)^2 = 9$

We went from (D) to (E) by multiplying out. Now go the other way. Move -5 to the right side: $x^2 - 4x = 5$.

Level 1 • threshold

Which number must you add to $x^2 - 4x$ to get $(x - 2)^2$?

This time we knew which square to aim for. How do you find the right square for an equation you have never seen, like $x^2 + 6x - 7 = 0$?

EXPLAIN

Completing the square

A square and a strip

Take $x^2 + 6x$. Draw x^2 as a square with side x , and $6x$ as a strip that is 6 wide and x long.

Split the strip

Cut the strip into two strips, each $3 \times x$. Put one beside the square and one below it. The area is still $x^2 + 6x$.

The 3 is **half of 6**.

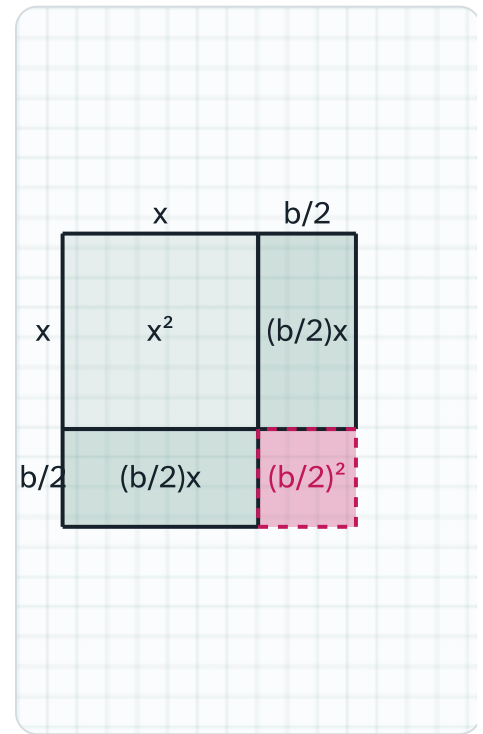
Fill the corner

One 3×3 corner is missing. Adding $3^2 = 9$ completes a square with side $x + 3$:

$$x^2 + 6x + 9 = (x + 3)^2.$$

The drawing only works for $x > 0$, but the identity holds for **every** x : multiply out $(x + 3)^2$ and you get $x^2 + 6x + 9$.

This is called **completing the square**.



The rule

For any number b , halve it and square the half:

$$x^2 + bx + \left(\frac{b}{2}\right)^2 = \left(x + \frac{b}{2}\right)^2.$$

Check it by multiplying out the right side:

$$\begin{aligned} \left(x + \frac{b}{2}\right)^2 &= x^2 + 2 \cdot x \cdot \frac{b}{2} + \left(\frac{b}{2}\right)^2 \\ &= x^2 + bx + \left(\frac{b}{2}\right)^2. \end{aligned}$$

- b negative: $x^2 - 4x + 4 = (x - 2)^2$. Half of -4 is -2 . The number added is its square, $(-2)^2 = 4$, which is positive. (The drawing has no strips of negative width, but the algebra works.)
- b odd: $x^2 + x + \frac{1}{4} = \left(x + \frac{1}{2}\right)^2$. Fractions are normal here.

A trinomial with any constant term becomes a square plus a number. Add $\left(\frac{b}{2}\right)^2$ and take it away again:

$$\begin{aligned} x^2 + 6x + 5 &= \underbrace{x^2 + 6x + 9}_{(x+3)^2} - 9 + 5 \\ &= (x + 3)^2 - 4. \end{aligned}$$

Solving: four moves

Now solve (E) without knowing (D) in advance. The column on the right explains each step. When we add a number, we add it to **both** sides.

$x^2 - 4x - 5 = 0$	$+ 5$
$x^2 - 4x = 5$	$+ \left(\frac{-4}{2}\right)^2 = 4$
$x^2 - 4x + 4 = 9$	left side is a square
$(x - 2)^2 = 9$	

Then $x - 2 = 3$ or $x - 2 = -3$, so $x = 5$ or $x = -1$.

The four moves for $ax^2 + bx + c = 0$:

1. If $a \neq 1$, divide both sides by a . This is allowed because $a \neq 0$, and it does not change the roots.
2. Move the constant term to the right side.
3. Add $\left(\frac{b}{2}\right)^2$ to **both** sides, where b is now the coefficient of x .
4. Write the left side as a square, $(x + p)^2 = d$, and solve it as in lesson 1.1.

Two roots, one, or none

Move 4 ends with $(x + p)^2 = d$, and d decides, as in lesson 1.1:

- $d > 0$. $x^2 - 2x - 1 = 0$ becomes $(x - 1)^2 = 2$. So $x - 1 = \sqrt{2}$ or $x - 1 = -\sqrt{2}$: **two roots**, $x = 1 + \sqrt{2}$ and $x = 1 - \sqrt{2}$. No amount of guessing whole numbers would find them.
- $d = 0$. $x^2 - 6x + 9 = 0$ becomes $(x - 3)^2 = 0$: **one root**, $x = 3$.
- $d < 0$. $x^2 + 2x + 5 = 0$ becomes $(x + 1)^2 = -4$: **no real roots**.

Two roots are often written together with the sign $\pm$ ("plus or minus"): $x = 1 \pm \sqrt{2}$ means $x = 1 + \sqrt{2}$ or $x = 1 - \sqrt{2}$.

Worked examples

Example 1 • $x^2 + 6x - 7 = 0$

MOVE C

$$x^2 + 6x = 7.$$

ADD $(6/2)^2 = 9$ TO BOTH SIDES

$$x^2 + 6x + 9 = 16.$$

WRITE THE SQUARE

$$(x + 3)^2 = 16.$$

SOLVE

$$x + 3 = 4 \text{ or } x + 3 = -4, \text{ so } x = 1 \text{ or } x = -7.$$

CHECK

$$1^2 + 6 \cdot 1 - 7 = 0 \checkmark \text{ and } (-7)^2 + 6 \cdot (-7) - 7 = 49 - 42 - 7 = 0 \checkmark.$$

Example 2 • An odd b: $x^2 + 5x + 4 = 0$

MOVE C

$$x^2 + 5x = -4.$$

ADD $(5/2)^2 = 25/4$ TO BOTH SIDES

$$x^2 + 5x + \frac{25}{4} = -4 + \frac{25}{4} = \frac{9}{4}.$$

WRITE THE SQUARE

$$\left(x + \frac{5}{2}\right)^2 = \frac{9}{4}.$$

SOLVE

$$x + \frac{5}{2} = \frac{3}{2} \text{ or } x + \frac{5}{2} = -\frac{3}{2}, \text{ so } x = -1 \text{ or } x = -4.$$

Example 3 • When $a \neq 1$: $4x^2 - 8x + 3 = 0$

DIVIDE BY $A = 4$

$$x^2 - 2x + \frac{3}{4} = 0. \text{ The rule needs the coefficient of } x^2 \text{ to be 1.}$$

MOVE C

$$x^2 - 2x = -\frac{3}{4}.$$

ADD $(-2/2)^2 = 1$ TO BOTH SIDES

$$x^2 - 2x + 1 = \frac{1}{4}, \text{ that is, } (x - 1)^2 = \frac{1}{4}.$$

SOLVE

$$x - 1 = \frac{1}{2} \text{ or } x - 1 = -\frac{1}{2}, \text{ so } x = \frac{3}{2} \text{ or } x = \frac{1}{2}.$$

CHECK

$$4 \cdot \frac{9}{4} - 8 \cdot \frac{3}{2} + 3 = 9 - 12 + 3 = 0 \checkmark \text{ and } 4 \cdot \frac{1}{4} - 8 \cdot \frac{1}{2} + 3 = 1 - 4 + 3 = 0 \checkmark.$$

SUMMARY

What to remember

- **Completing the square:** halve the coefficient of x and square the half: $x^2 + bx + \left(\frac{b}{2}\right)^2 = \left(x + \frac{b}{2}\right)^2$.
- Any $x^2 + bx + c$ is a square plus a number: $x^2 + 6x + 5 = (x + 3)^2 - 4$.
- **The four moves** for $ax^2 + bx + c = 0$:
 1. if $a \neq 1$, divide both sides by a ;
 2. move the constant term to the right;
 3. add $\left(\frac{b}{2}\right)^2$ to **both** sides;
 4. write $(x + p)^2 = d$ and solve it.
- $d > 0$: two roots, $x = -p \pm \sqrt{d}$; $d = 0$: one root, $x = -p$; $d < 0$: no real roots.
- Fractions are normal when b is odd or $a \neq 1$. Check the roots by substituting them.

NEXT LESSON

We made the same four moves every time, only with different numbers. What do we get if we make them once, with the letters a , b and c ?

[Lesson 1.3 · The quadratic formula →](#)